

Theory 1. Convergence speed and second differential.

Task 1 (10/7 pts) For each of the following sequences $\{r_k\}$ find out its convergence speed (linear, sublinear, superlinear). In case of superlinear rate find out additionally whether the sequence has quadratic convergence rate.

(a) $r_k = (0.99)^{k^2}$ (b) $r_k = (0.99)^{2^k}$ (c) $r_k = 1/k!$

(d) $r_k = 1/k^k$ (e) $r_k = \begin{cases} (0.99)^{2^k}, & \text{if } k \text{ is even,} \\ r_{k-1}/k, & \text{otherwise.} \end{cases}$

Task 2 (10/7 pts) Suppose that $\{r_k\}$ – the sequence of non-negative values given by $r_{k+1} = Mr_k^2$, where $M > 0$, $r_0 \geq 0$. Find necessary and sufficient condition for M and r_0 , when the sequence converges to zero. What would be the convergence rate in this case?

Task 3 (40/7 pts) For each of the following function find out the second differential $d^2f(\mathbf{x})[d\mathbf{x}, d\mathbf{x}]$ and show that this differential has constant sign. Find out also whether this sign is strict for all $d\mathbf{x} \neq \mathbf{0}$ or not:

- (a) $f(\mathbf{x}) = \ln\left(-\frac{1}{2}\mathbf{x}^T A \mathbf{x} - \mathbf{b}^T \mathbf{x} - c\right)$, where A is non-negatively definite matrix and the function is considered only for those values of $\mathbf{x}$, where the expression under logarithm is strictly positive.
- (b) $f(\mathbf{x}) = (\sum_{i=1}^n x_i^p)^{1/p}$, where $p < 1, p \neq 0, x_i > 0 \forall i$.
- (c) $f(X) = \text{tr}(X^{-1}A)$, where X is positive definite matrix and A is non-negative definite matrix.
- (d) $f(X) = (\det(X))^{1/n}$, where X is positive definite matrix.

Hint: use Cauchy-Schwarz inequality. Positive or non-negative definite matrix is always supposed to be also symmetric.

Task 4 (10/7 pts) Let's consider the function $f(\mathbf{x}) = (\mathbf{x}^T A \mathbf{x})/\|\mathbf{x}\|^2$, where A – symmetric matrix and $\mathbf{x} \neq \mathbf{0}$. Show that all stationary points of this function are determined by eigenvectors of the matrix A . Find the Hessian $\nabla^2 f(\mathbf{x})$ for eigenvectors $\mathbf{x}$. Determine the type of stationarity (local minima, local maxima, saddle point).